

Department of Mechanical and Production Engineering

Aarhus University

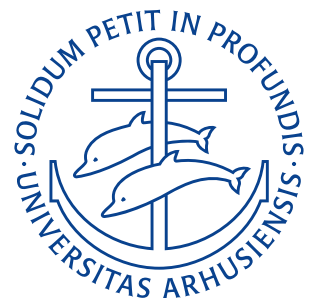
Take-Home Assignment Lec. 18–21

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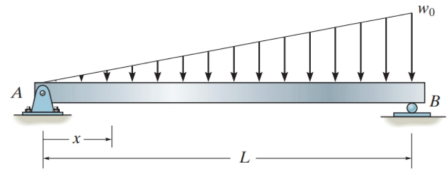
Course: Statics & Strength of Materials — 290221U006

Date: 9. November 2025

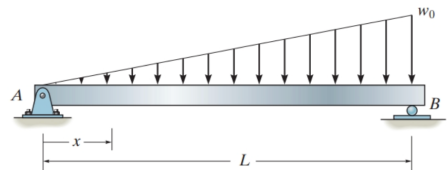


Problem 16.17

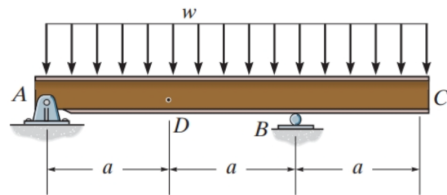
Determine the maximum deflection of the beam. EI is constant.

**Problem 16.18**

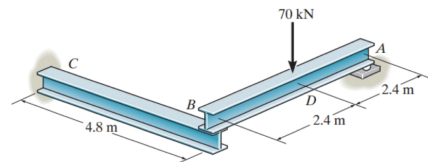
Determine the maximum slope of the beam. EI is constant.

**Problem 16.36**

Determine the slope at A and the deflection at point D of the overhang beam. EI is constant.

**Problem 16.46**

The assembly consists of a cantilevered beam CB and a simply supported beam AB . If each beam is made of A-36 steel and has a moment of inertia about its principal axis of $I_x = 50 \cdot 10^6 \text{ mm}^4$, determine the displacement at the center D of beam BA .

**Problem R16.7**

Using the method of superposition, determine the displacement at C of beam AB. The beams are made of wood having a modulus of elasticity of $E = 10 \text{ GPa}$.

